PTCMGH: Overall Survival Analysis

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Data

CALGB 40101 (NCT00041119) is a Phase III trial finding that 4 cycles of chemotherapy are non-inferior to 6 cycles for early-stage breast cancer (0-3 positive nodes).

The following R code illustrates how to analyse overall survival in this data set using the `PTCMGH` R package.

Data preparation

```
rm(list=ls())

#library(devtools)
#install_github("FJRubio67/HazReg")
library(HazReg)
#library(devtools)
#install_github("FJRubio67/PTCMGH")
library(PTCMGH)
library(survival)
library(knitr)
library(haven)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
# Read the raw data set
df <- read_sas("all_finalb.sas7bdat")
head(df)
```

```
## # A tibble: 6 x 37
##   mask_id SSTAT ETHNIC_ID scase GROUP_ID RACE_ID indrx stra1 stra2 stra3
##     <dbl> <dbl>     <dbl> <dbl>    <dbl>   <dbl> <dbl> <dbl> <dbl> <dbl>
## 1       1     7         1    10       37       1     4     1     1     2
## 2       2     7         2    10       37       1     1     2     1     2
## 3       3     7         9    10        1       3     1     1     2     3
## 4       4     7         2    10       37       1     1     1     1     3
## 5       5     7         2    10        1       1     3     2     1     2
## 6       6     8         2    11       37       1     1     2     1     3
## # i 27 more variables: survstat <dbl>, elig <dbl>, OH002 <dbl>, OH003 <dbl>,
## #   OH004 <dbl>, OH005 <dbl>, OH011 <dbl>, OH012 <dbl>, OH013 <dbl>,
## #   OH014 <dbl>, OH016 <dbl>, OH027 <dbl>, OH028 <dbl>, OH032 <dbl>,
## #   OH036 <dbl>, OH037 <dbl>, tsize <dbl>, cod <dbl>, agecat <dbl>,
## #   preamend <dbl>, survmos <dbl>, event <dbl>, dfsstat <dbl>, dfsmos <dbl>,
## #   agent <dbl>, length <dbl>, num_pos_nodes <chr>
```

```r
# Data
df_raw <- read_sas("all_finalb.sas7bdat")

df <- df_raw %>%
  # Restrict to eligible patients only
  filter(elig == 2) %>%
  mutate(
    # --- Outcome variables ---
    os_time   = survmos,
    os_event  = survstat,       # 0=alive, 1=dead
    dfs_time  = dfsmos,
    dfs_event = dfsstat,        # 0=no event, 1=event

    # --- Treatment (primary exposures) ---
    agent  = factor(agent,  levels = c(0,1), labels = c("CA", "T")),
    length = factor(length, levels = c(0,1), labels = c("4 cycles", "6 cycles")),

    # --- Stratification / baseline covariates ---
    menopause = factor(stra1, levels = c(1,2), labels = c("Pre", "Post")),
    receptor  = factor(stra2, levels = c(1,2), labels = c("Pos/Unk", "Neg")),
    her2      = factor(stra3, levels = c(1,2,3), labels = c("Positive","Negative","Unknown")),

    # --- Tumour characteristics ---
    er_status   = factor(OH003, levels = c(1,2), labels = c("Negative","Positive")),
    pgr_status  = factor(OH004, levels = c(1,2), labels = c("Negative","Positive")),
    grade       = factor(OH005, levels = c(1,2,3), labels = c("Low","Intermediate","High")),
    tumor_size  = factor(tsize, levels = c(1,2,3), labels = c("<2cm","2-5cm",">5cm")),
    num_pos_nodes = num_pos_nodes,   # keep as-is or convert to numeric (note ">3" category)
```

```r
    # --- Demographics ---
    age_cat = factor(agecat, levels = 1:6,
                     labels = c("20-29","30-39","40-49","50-59","60-69","70+")),
    race    = factor(RACE_ID, levels = c(1,3,4,5,99),
                     labels = c("White","Black","Asian","Other","Unknown")),
    ethnicity = factor(ETHNIC_ID, levels = c(1,2,9),
                       labels = c("Hispanic","Non-Hispanic","Unknown")),

    # --- Surgery / staging ---
    surgery_type = factor(OH028, levels = c(1,2), labels = c("Lumpectomy","Mastectomy")),
    sentinel_bx  = factor(OH032, levels = c(1,2), labels = c("No","Yes")),
    axillary_diss = factor(OH037, levels = c(1,2), labels = c("No","Yes")),

    # --- Amendment (important! 6-cycle arms were later closed) ---
    preamend = factor(preamend, levels = c(0,1), labels = c("Post-amend","Pre-amend"))
  )

dim(df)
```

```
## [1] 3609   54
```

```r
# Complete cases for variables of interest
dfc <- data.frame(agent = df$agent, length = df$length, menopause = df$menopause,
                  receptor = df$receptor, tumor_size = df$tumor_size, grade = df$grade,
                  num_pos_nodes = as.numeric(df$num_pos_nodes), age_cat = df$age_cat,
                  indrx = df$indrx, her2 = df$her2,
                  os_time = df$os_time/12, os_event = df$os_event)
```

```
## Warning in data.frame(agent = df$agent, length = df$length, menopause =
## df$menopause, : NAs introduced by coercion
```

```r
dfc <- dfc[complete.cases(dfc),]
dim(dfc)
```

```
## [1] 3560   12
```

```r
# Build the design matrix from the Cox model formula
des0 <- model.matrix(
  ~ factor(agent) + factor(length) + factor(menopause) +
    factor(receptor) + factor(her2) + factor(tumor_size) +
    factor(grade),
  data = dfc
)

# View the first few rows
head(des0)
```

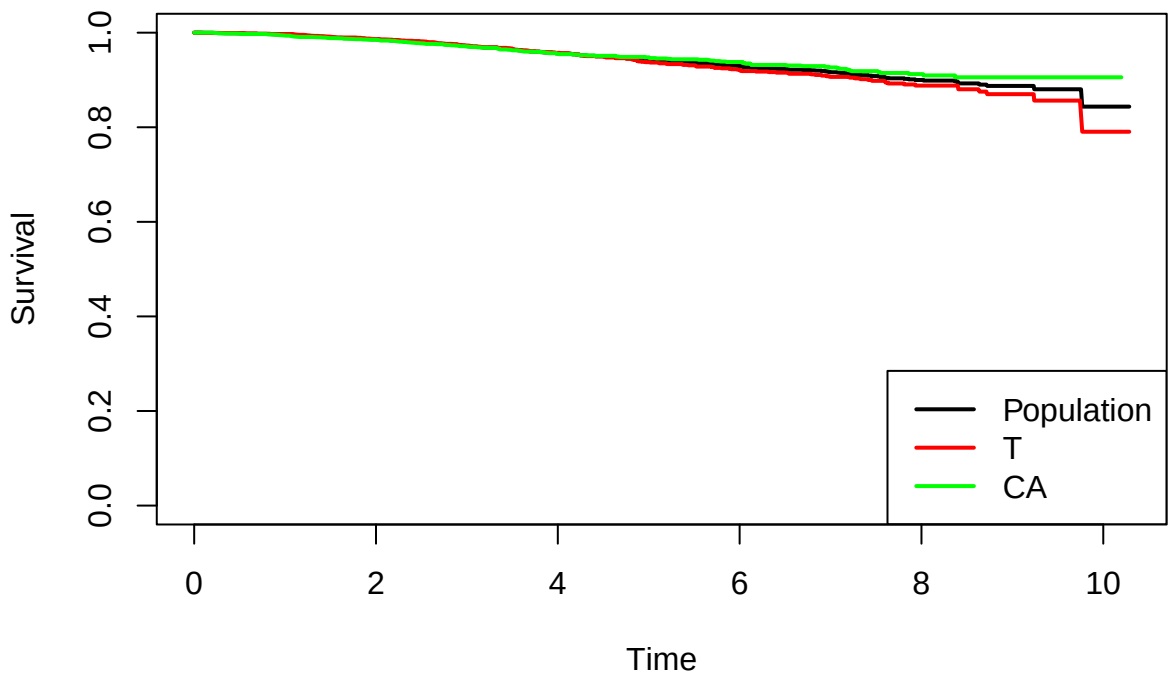
```
##   (Intercept) factor(agent)T factor(length)6 cycles factor(menopause)Post
## 1           1              1                      1                     0
## 2           1              0                      0                     1
## 3           1              0                      0                     0
## 4           1              0                      0                     0
## 5           1              1                      0                     1
## 6           1              0                      0                     1
##   factor(receptor)Neg factor(her2)Negative factor(her2)Unknown
## 1                   0                    1                    0
```

```
## 2                   0                    1                   0
## 3                   1                    0                   1
## 4                   0                    0                   1
## 5                   0                    1                   0
## 6                   0                    0                   1
##   factor(tumor_size)2-5cm factor(tumor_size)>5cm factor(grade)Intermediate
## 1                       0                      0                         1
## 2                       0                      0                         1
## 3                       1                      0                         0
## 4                       1                      0                         0
## 5                       0                      0                         1
## 6                       1                      0                         1
##   factor(grade)High
## 1                 0
## 2                 0
## 3                 1
## 4                 1
## 5                 0
## 6                 0
```

```r
dim(des0)
```

```
## [1] 3560   11
```

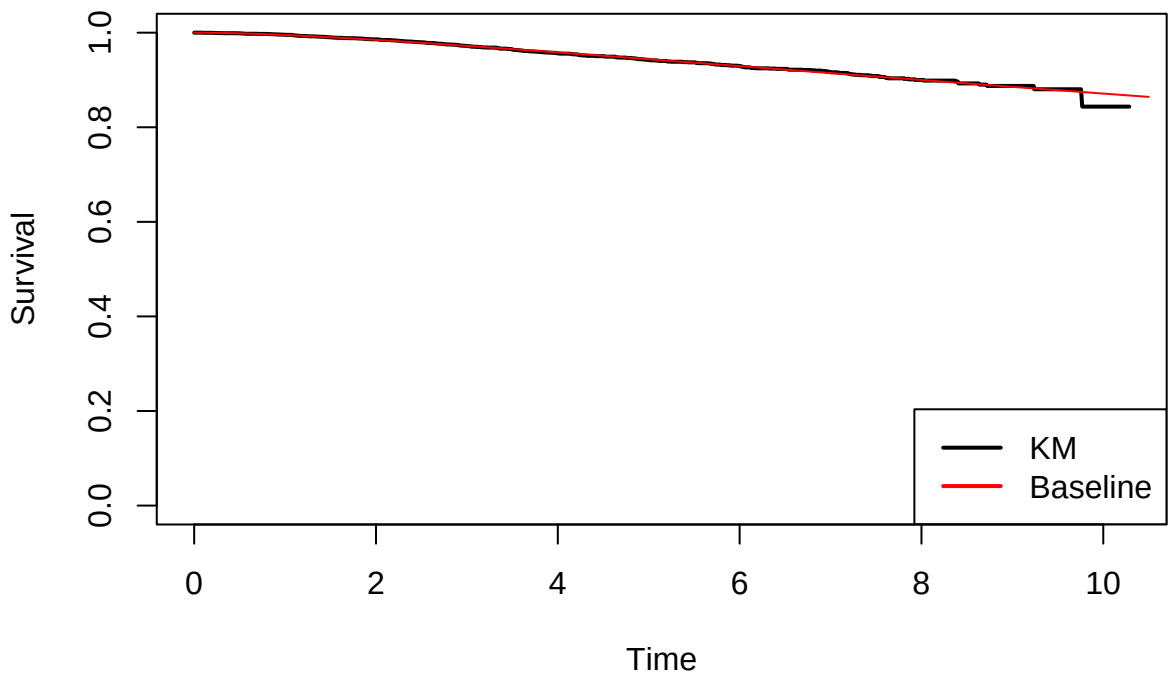
```r
nc <- nrow(dfc)


# Survival times
times <- dfc$os_time
# Status
status <- dfc$os_event

# Kaplan-Meier estimator for the overall survival times
km <- survfit(Surv(times, status) ~ 1)
plot(km$time, km$surv, type = "l", col = "black", lwd = 2, lty = 1,
     ylim = c(0,1), xlab = "Time", ylab = "Survival")


# Kaplan-Meier estimator for the overall survival times: T (0) vs CA (1)
ind_t <- which(dfc$agent == "T")
ind_ca <- which(dfc$agent == "CA")

km_t <- survfit(Surv(times[ind_t], status[ind_t]) ~ 1)
km_ca <- survfit(Surv(times[ind_ca], status[ind_ca]) ~ 1)

plot(km$time, km$surv, type = "l", col = "black", lwd = 2, lty = 1,
     ylim = c(0,1), xlab = "Time", ylab = "Survival")
points(km_t$time, km_t$surv, type = "l", col = "red", lwd = 2, lty = 1,
     ylim = c(0,1), xlab = "Time", ylab = "Survival")
points(km_ca$time, km_ca$surv, type = "l", col = "green", lwd = 2, lty = 1,
     ylim = c(0,1), xlab = "Time", ylab = "Survival")
legend("bottomright", legend = c("Population","T","CA"), col = c("black","red","green"),
       lwd = c(2,2,2))
```



Model fit

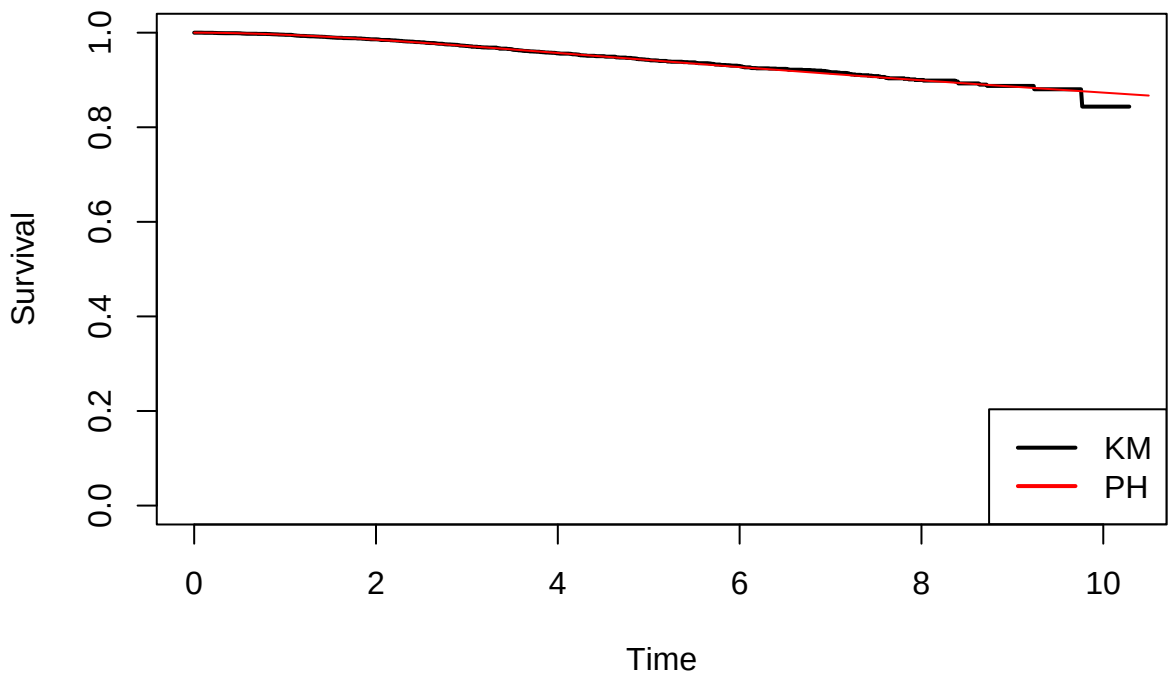
Baseline

```
# Model fitting
OPT <- PTCMMLE(init = c(0,0,0),
                     times = times,
                     status = status,
                     hstr = "baseline",
                     dist = "LogNormal",
                     des_theta = NULL,
                     des_t = NULL,
                     des_h = NULL,
                     des = NULL,
                     method = "nlminb",
                     maxit = 10000)

# MLE
MLE <- c(OPT$OPT$par[1],exp(OPT$OPT$par[2]),exp(OPT$OPT$par[3]))

# Fitted survival function
spt <- Vectorize(function(t) exp(- MLE[3]*(1-exp(-chlnorm(t,MLE[1],MLE[2])))) )

plot(km$time, km$surv, type = "l", col = "black", lwd = 2, lty = 1,
     ylim = c(0,1), xlab = "Time", ylab = "Survival")
curve(spt,0,10.5, col = "red", add= T)
legend("bottomright", legend = c("KM","Baseline"), col = c("black","red"),
       lwd = c(2,2))
```



PH

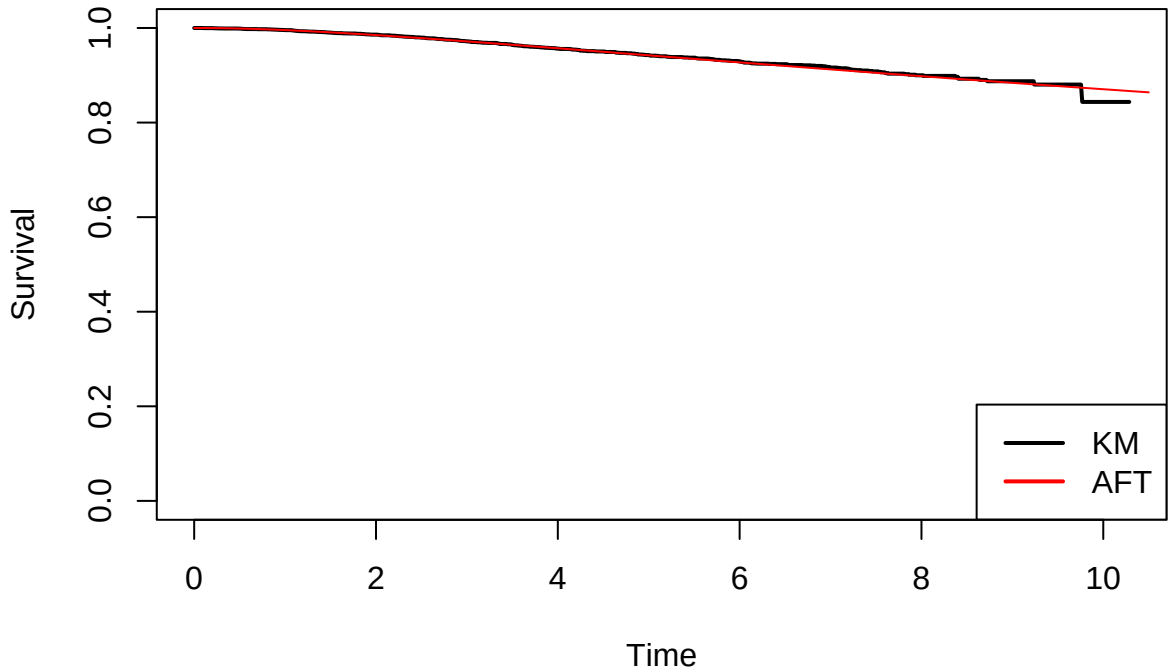
```
# Model fitting
OPT_PH <- PTCMMLE(init = rep(0,2*ncol(des0)+1),
                times = times,
                status = status,
                hstr = "PH",
                dist = "LogNormal",
                des_theta = des0,
                des_t = NULL,
                des_h = NULL,
                des = des0[,-1],
                method = "nlminb",
                maxit = 10000)

# MLE
MLE_PH <- c(OPT_PH$OPT$par[1],exp(OPT_PH$OPT$par[2]),OPT_PH$OPT$par[-c(1:2)])

# Population survival function
spt_ph <- Vectorize(function(t){

  theta_i <- as.vector(exp(des0 %*% MLE_PH[3:(2+ncol(des0))]))
  F_i <- 1 - exp(-chlnorm(t,MLE_PH[1],MLE_PH[2])*exp(des0[,-1]%*%MLE_PH[(3+ncol(des0)):(2*ncol(des0)+1)]
  survs <- exp(-theta_i*F_i)
  return(mean(survs))
  })

plot(km$time, km$surv, type = "l", col = "black", lwd = 2, lty = 1,
     ylim = c(0,1), xlab = "Time", ylab = "Survival")
curve(spt_ph,0,10.5, col = "red", add= T)
legend("bottomright", legend = c("KM","PH"), col = c("black","red"),
       lwd = c(2,2))
```



AFT

```
# Model fitting
OPT_AFT <- PTCMMLE(init = rep(0,2*ncol(des0)-1+2),
                   times = times,
                   status = status,
                   hstr = "AFT",
                   dist = "LogNormal",
                   des_theta = des0,
                   des_t = NULL,
                   des_h = NULL,
                   des = des0[,-1],
                   method = "nlminb",
                   maxit = 10000)

# MLE
MLE_AFT <- c(OPT_AFT$OPT$par[1],exp(OPT_AFT$OPT$par[2]),OPT_AFT$OPT$par[-c(1:2)])

# Population survival function
spt_aft <- Vectorize(function(t){

  theta_i <- as.vector(exp(des0 %*% MLE_AFT[3:(2+ncol(des0))]))
  F_i <- 1 - exp(-chlnorm(as.vector(t*exp(des0[,-1]%*%MLE_AFT[(3+ncol(des0)):(2*ncol(des0)+1)])),MLE_AF
  survs <- exp(-theta_i*F_i)
  return(mean(survs))
})


plot(km$time, km$surv, type = "l", col = "black", lwd = 2, lty = 1,
     ylim = c(0,1), xlab = "Time", ylab = "Survival")
curve(spt_aft,0,10.5, col = "red", add= T)
legend("bottomright", legend = c("KM","AFT"), col = c("black","red"),
```

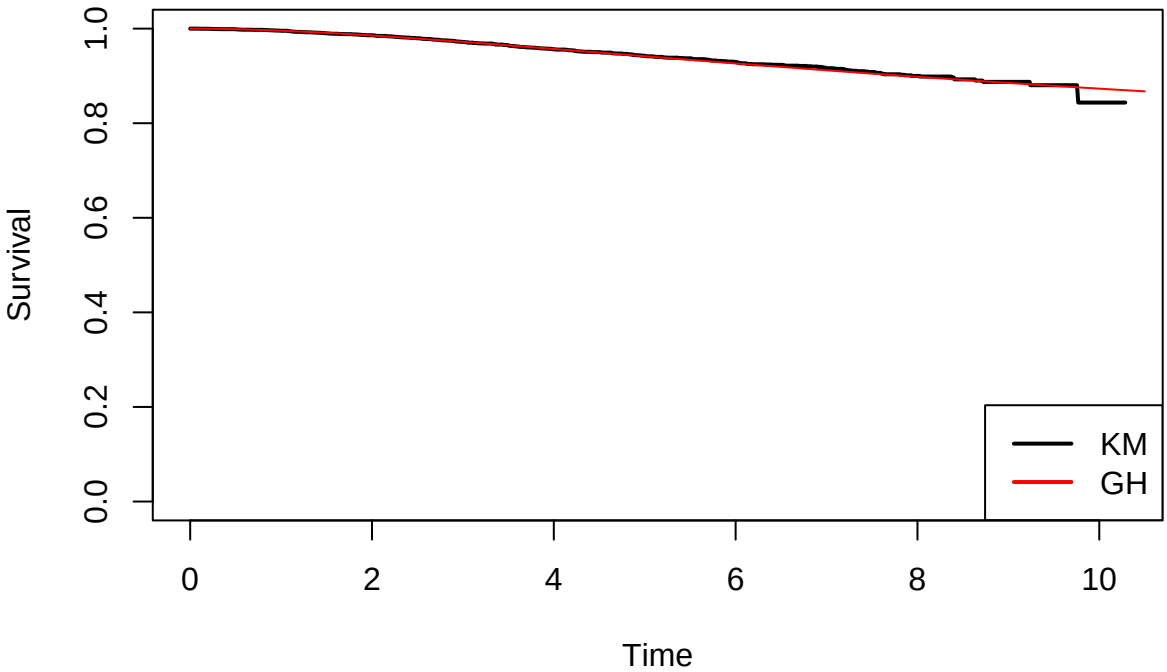
```
       lwd = c(2,2))
```



General Hazards

```
# Model fitting
OPT_GH <- PTCMMLE(init = rep(0,3*ncol(des0)),
                  times = times,
                  status = status,
                  hstr = "GH",
                  dist = "LogNormal",
                  des_theta = des0,
                  des_t = des0[,-1],
                  des_h = des0[,-1],
                  des = NULL,
                  method = "nlminb",
                  maxit = 10000)

# MLE
MLE_GH <- c(OPT_GH$OPT$par[1],exp(OPT_GH$OPT$par[2]),OPT_GH$OPT$par[-c(1:2)])


# Population survival function
spt_gh <- Vectorize(function(t){

  theta_i <- as.vector(exp(des0 %*% MLE_GH[3:(2+ncol(des0))]))
  F_i <- 1 - exp(-chlnorm(as.vector(t*as.vector(exp(des0[,-1]%*%MLE_GH[(3+ncol(des0)):(2*ncol(des0)+1)]
                  as.vector(exp(des0[,-1]%*%MLE_GH[(2*ncol(des0)+2):(3*ncol(des0))] - des0[,-1]%*%MLE_
  survs <- exp(-theta_i*F_i)
  return(mean(survs))
})

plot(km$time, km$surv, type = "l", col = "black", lwd = 2, lty = 1,
```

```
     ylim = c(0,1), xlab = "Time", ylab = "Survival")
curve(spt_gh,0,10.5, col = "red", add= T)
legend("bottomright", legend = c("KM","GH"), col = c("black","red"),
       lwd = c(2,2))
```



Model comparison

```
# Akaike information criterion
AIC <- 2*OPT$OPT$objective + 2*length(OPT$OPT$par)
AIC_PH <- 2*OPT_PH$OPT$objective + 2*length(OPT_PH$OPT$par)
AIC_AFT <- 2*OPT_AFT$OPT$objective + 2*length(OPT_AFT$OPT$par)
AIC_GH <- 2*OPT_GH$OPT$objective + 2*length(OPT_GH$OPT$par)

# AICs: baseline, PH, AFT, AH, GH
AICs <- c(AIC, AIC_PH, AIC_AFT, AIC_GH)

AICs
```

```
## [1] 2775.687 2705.955 2708.119 2710.414
```

```
which.min(AICs)
```

```
## [1] 2
```

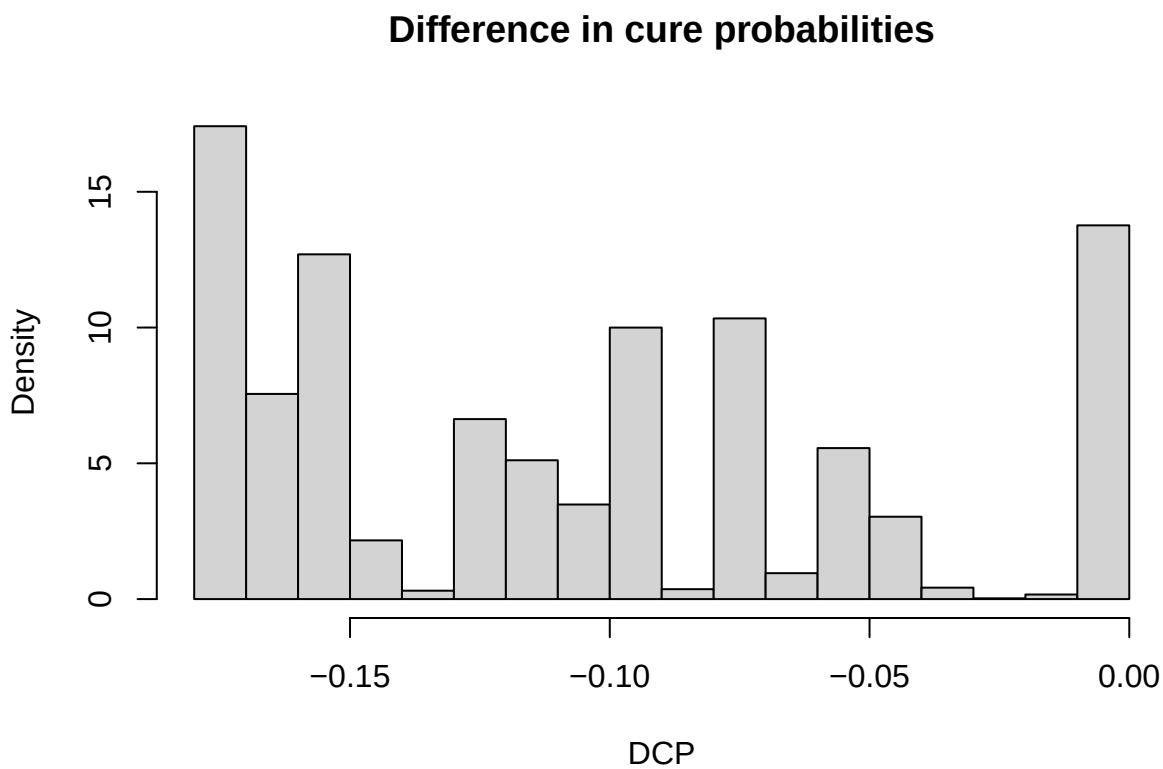
Estimands: PH model

Incidence

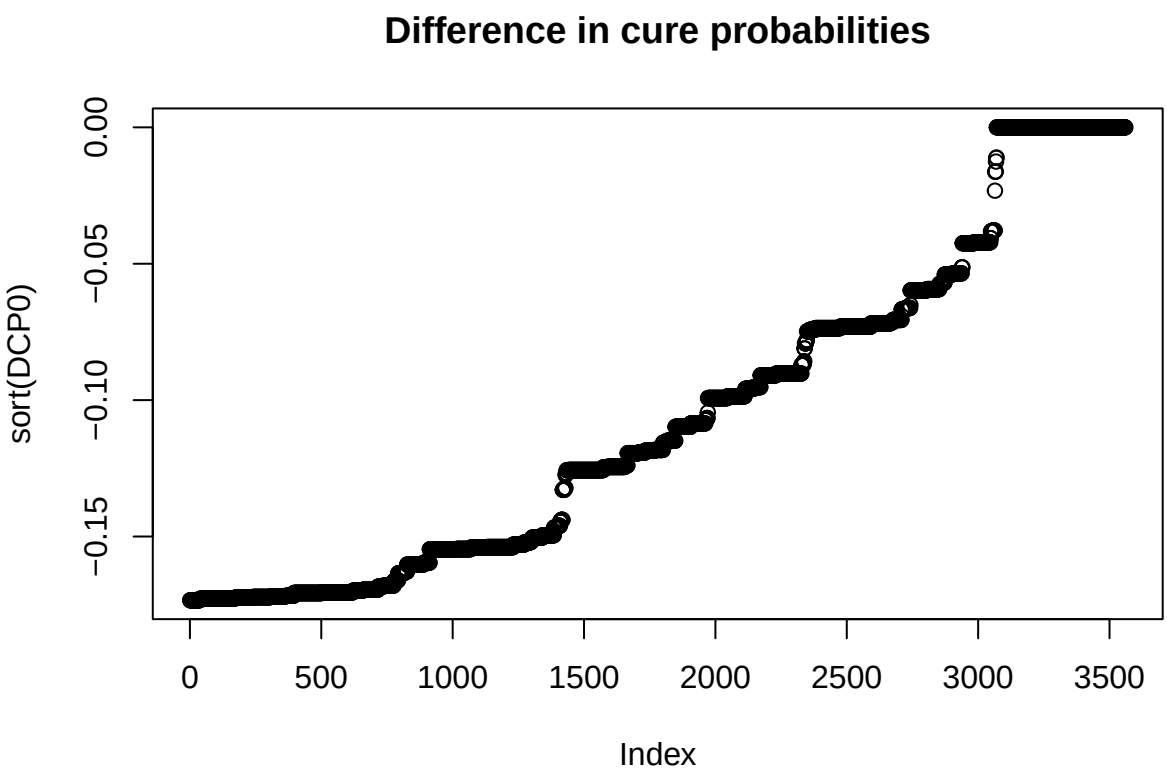
Difference in cure probabilities

```
# x: covariate values
# index: index of x corresponding to treatment (agent in this case)
diff_CP <- function(x, index){
```

```
  x0 <- as.matrix(x)
  x0[,index] = 0
  x1 <- as.matrix(x)
  x1[,index] = 1
  theta_i0 <- as.vector(exp(x0 %*% MLE_PH[3:(2+ncol(x0))]))
  theta_i1 <- as.vector(exp(x1 %*% MLE_PH[3:(2+ncol(x1))]))

  dif_theta <- exp(-theta_i1) - exp(-theta_i0)
  return(as.vector(dif_theta))
}

DCP0 <- diff_CP(des0,2)

hist(DCP0, main = "Difference in cure probabilities",
     xlab = "DCP", probability = TRUE)
```




```
plot(sort(DCP0), main = "Difference in cure probabilities")
```




```r
# Average treatment effect (ATE): Cure probabilities
ATE_CP <- mean(DCP0)
ATE_CP
```

```
## [1] -0.1063553
```

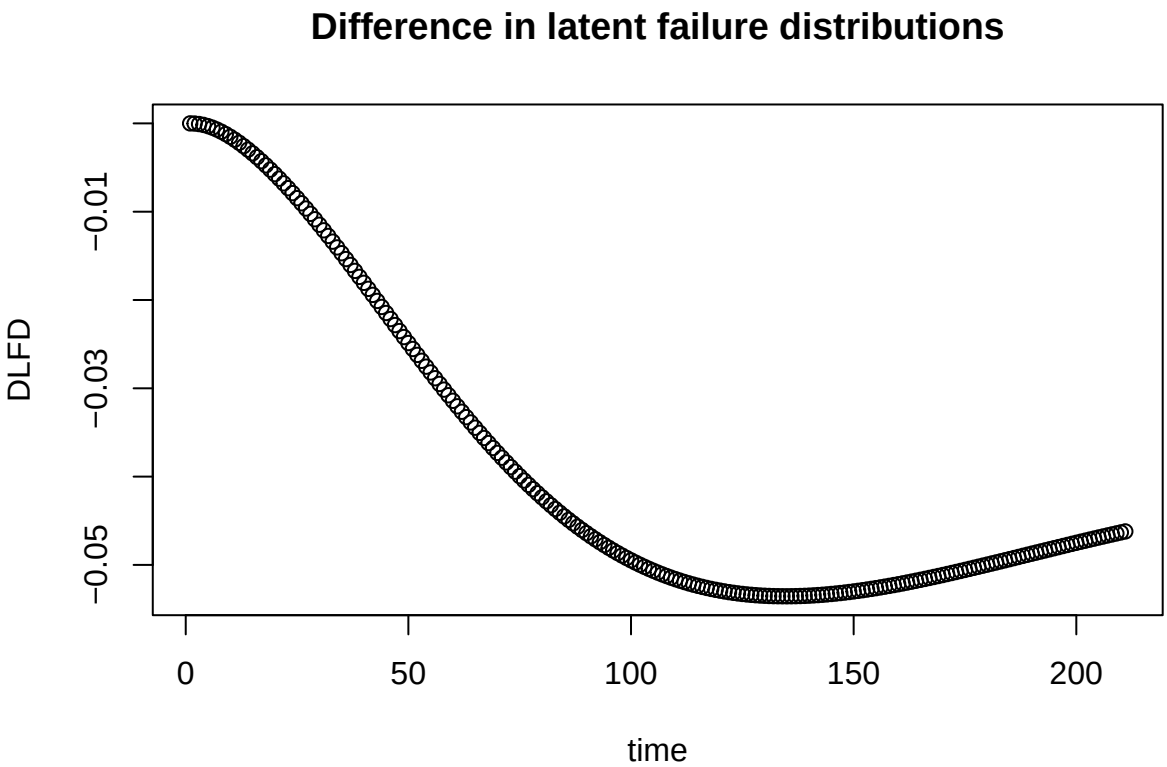
```r
summary(DCP0)
```

```
##     Min.  1st Qu.   Median     Mean  3rd Qu.     Max.
## -0.17336 -0.15962 -0.11825 -0.10636 -0.07141  0.00000
```

Difference in latent failure distributions

```r
# x: covariate values
# index: index of x corresponding to treatment (agent in this case)
diff_LFD <- function(t, x, index){
  x0 <- as.matrix(x)
  x0[,index] = 0
  x1 <- as.matrix(x)
  x1[,index] = 1
  F_i0 <- 1 - exp(-chlnorm(t,MLE_PH[1],MLE_PH[2])*exp(x0[,-1]%*%MLE_PH[(3+ncol(x0)):(2*ncol(x0)+1)]))
  F_i1 <- 1 - exp(-chlnorm(t,MLE_PH[1],MLE_PH[2])*exp(x1[,-1]%*%MLE_PH[(3+ncol(x1)):(2*ncol(x1)+1)]))

  dif_lfd <- F_i1 - F_i0
  return(as.vector(dif_lfd))
}

ts <- seq(0,10.5, by = 0.05)

DLFD0 <- matrix(0, nrow = nc, ncol = length(ts))
```

```r
for(i in 1:length(ts)) DLFD0[,i] <- diff_LFD(ts[i], des0, 2)

# Average treatment effect (ATE): latent failure distributions
plot(colMeans(DLFD0), main = "Difference in latent failure distributions",
     xlab = "time", ylab = "DLFD")
```



References